

A-Grid: SmartGrid Monitoring and Fault Detection System using AI and PoA² Blockchain

1. Background and Problem Statement

Modern power systems face critical challenges in energy monitoring, fault detection, and data trust losses over \$96B annually to Non-Technical Loss (NTL) and undetected faults. Current monitoring solutions suffer from:

- Data Integrity Problem (Oracle Issue): 90% of Smart meters data can be manipulated before reaching the system.
- Response Latency: Centralized systems take minutes to identify faults that cause damage in milliseconds.
- Lack of Trusted Records: No tamper-proof system for auditing faults and energy data
- Scalability Issues: Traditional systems struggle with real-time data processing

2. Proposed Solution: A-Grid Framework

A-Grid introduces a hybrid AI-blockchain smart grid monitoring system that ensures:

1. Data Trust Layer

- Software-based validation + optional hardware trust (TEE-ready design)
- Data is verified before blockchain submission
- Future-ready for hardware security integration

2. Edge Intelligence Layer (Core Innovation)

- AI-based anomaly detection (Isolation Forest / LSTM)
- Real-time fault detection at the edge (milliseconds level)
- Detects:
 - Voltage anomalies
 - Energy theft patterns
 - Equipment faults

3. Blockchain Trust Layer

- Uses **PoA² consensus via Purechain Open Cloud**
- Records:
 - Verified data (hash)
 - Fault events
 - System state

- Ensures:
 - Immutability
 - Traceability
 - Low latency and high TPS

4. System Architecture

4.1 IoT / Data Layer

- Smart meters (or simulated devices for prototype)
- Measures:
 - Energy production
 - Energy consumption
 - Voltage / current

4.2 Edge & Middleware Layer

- Built using Node.js / Python
- Performs:
 - Data validation
 - AI-based fault detection
 - Data filtering

4.3 Storage Layer

- **IPFS (off-chain):**
 - Stores full sensor data
- **Blockchain (on-chain):**
 - Stores hash + metadata

4.4 Blockchain Layer

- Purechain PoA² (via API)
- Handles:
 - Transaction validation
 - Consensus
 - Immutable recording

4.5 Application Layer

- Dashboard (optional for prototype)
- Displays:
 - Energy usage
 - Fault alerts

- Historical data

3. Technical Metrics & Performance Analysis

Metric	Target
Fault Detection Latency	< 100 ms (Edge AI)
Blockchain Finality	< 2–3 seconds
Throughput	500–2000 TPS (PoA ²)
Storage Reduction	> 70% (IPFS hybrid)

4. AI-Based Fault Detection (Practical Version)

Phase 1 (Prototype):

- Isolation Forest (fast, lightweight)
- Phase 2 (Research Extension):
- LSTM for time-series prediction

Capabilities:

- Detect anomalies in real-time
- Identify abnormal consumption patterns
- Improve detection accuracy over time

5. Implementation Roadmap

Phase 1: Prototype Development

- Set up backend (Node.js + Python AI)
- Integrate PoA² API
- Simulate smart meter data

Phase 2: System Integration

- Connect AI + backend + blockchain
- Integrate IPFS storage
- Implement validation layer

Phase 3: Testing & Evaluation

- Measure:
- Latency

- TPS
- Detection accuracy

Phase 4: Optimization

- Improve performance
- Add dashboard
- Prepare demo

6. Innovation and Contribution

A-Grid introduces:

- AI-based real-time fault detection
- Multi-layer validation (before blockchain)
- Hybrid storage (IPFS + blockchain)
- Efficient consensus using PoA²
- Cyber-physical smart grid monitoring

7. Commercial Viability & Impact (The ROI)

- Utility Benefit: Expected 15–20% reduction in Non-Technical Loss within the first 12 months.
- Insurance Value: "Forensic-as-a-Service" model where insurers pay for access to the immutable fault ledger to verify claims.
- Incentive Model: A P2P "Energy Trust" score. Prosumers with high scores (verified green production) receive lower transaction fees on the network.
- Support future smart grid infrastructure.

8. References

IEEE 2030.5: The standard for Smart Energy Profile (SEP) communications.

NIST IR 7628: Guidelines for Smart Grid Cybersecurity.

IEC 61850: The international standard for communication protocols in substations.

ISO/IEC 27001: For the information security management system of your blockchain nodes.